A FINITE LOCALLY FREE MORPHISM OF DEGREE D CARRIES A CANONICAL NORM OF DEGREE D COMPATIBLE WITH BASE CHANGE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-finite-locally-free-has-norm is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 3788-3793 and proof span 3795-3823 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof's references to 'property (1)' and 'property (2)' point at the defining properties of a norm of degree d, which are part of the surrounding definition in the source; they are the author's wording and are preserved.

Allowed standard prerequisites (editorial).

- A finite locally free morphism has a pushforward of the structure sheaf which is finite locally free of the given rank, so that after passing to a suitable affine open a basis can be chosen. Used at proof lines 3795-3797. Produces the identification of the pushforward with a free module of rank d that carries the matrix of multiplication by a section.
- Determinants of matrices: the determinant is independent of the chosen basis and the determinant of a diagonal matrix is the product of its entries. Used at proof lines 3803-3806 and 3810-3812. Makes the norm well defined and independent of the local trivialization, and gives the degree property.
- Over the spectrum of a field a finite dimensional algebra of dimension d admits a map to a field evaluating a vanishing section at zero, and a non-surjective endomorphism has determinant zero. Used at proof lines 3813-3823. Supplies the vanishing criterion for the norm after base change to a field.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite locally free morphism of degree $d \geq 1$. Then there exists a canonical norm of degree d whose formation commutes with arbitrary base change.*

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Proof. Let $V \subset Y$ be an affine open such that $(\pi_*\mathcal{O}_X)|_V$ is finite free of rank d. Choosing a basis we obtain an isomorphism

$$\mathcal{O}_V^{\oplus d} \cong (\pi_*\mathcal{O}_X)|_V$$

For every $f \in \pi_*\mathcal{O}_X(V) = \mathcal{O}_X(\pi^{-1}(V))$ multiplication by f defines a $\mathcal{O}_V$-linear endomorphism m_f of the displayed free vector bundle. Thus we get a $d \times d$ matrix $M_f \in \mathrm{Mat}(d \times d, \mathcal{O}_Y(V))$ and we can set

$$\mathrm{Norm}_\pi(f) = \det(M_f)$$

Since the determinant of a matrix is independent of the choice of the basis chosen we see that this is well defined which also means that this construction will glue to a global map as desired. Compatibility with base change is straightforward from the construction.

Property (1) follows from the fact that the determinant of a $d \times d$ diagonal matrix with entries $g, g, \ldots, g$ is g^d. To see property (2) we may base change and assume that Y is the spectrum of a field k. Then $X = \mathrm{Spec}(A)$ with A a k-algebra with $\dim_k(A) = d$. If there exists an $x \in X$ such that $f \in A$ vanishes at x, then there exists a map $A \to \kappa$ into a field such that f maps to zero in κ. Then $f : A \to A$ cannot be surjective, hence $\det(f : A \to A) = 0$ as desired. $\square$